PRACTICAL / TUTORIAL NO. 15

AUTOMATED LOGIN TESTING

STUDENT DETAILS	
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Program	BSc CS IT

Subject: Software Testing / Selenium Web Automation

Application Under Test: SauceDemo

Contents: practical workbook tables, test scenarios, element identification, validation reports, execution summary, screenshot checklist, and visual reference pages.

Implementation Details

TUTORIAL 15 - AUTOMATED LOGIN TESTING	
Parameter	Details
Aim	Automate login functionality, authentication validation, assertions, and exception handling using Selenium WebDriver with Python.
Project Name	Automated Login Testing
Application	SauceDemo Web Application
Module	Login / Authentication
Programming Language	Python
IDE / Tool	VS Code, Selenium WebDriver
Test Data	Valid: standard_user / secret_sauce; invalid and empty credentials
Target URL	https://www.saucedemo.com/

Environment Setup

Parameter	Details
Python Version	Python 3.x
Selenium Version	Install using pip
Browser	Google Chrome
WebDriver	Selenium Manager / ChromeDriver
IDE	VS Code
Install Command	python -m pip install selenium
Run Command	python tutorial15_login_testing.py

Login Test Scenarios

Test Case ID	Scenario	Test Data	Expected Result	Actual Result	Status
TC-01	Valid Login	standard_user / secret_sauce	Products page opens	Record after execution	Pending
TC-02	Invalid Password	standard_user / wrong_password	Error message appears	Record after execution	Pending
TC-03	Invalid Username	wrong_user / secret_sauce	Error message appears	Record after execution	Pending
TC-04	Empty Credentials	Blank username and password	Validation error appears	Record after execution	Pending
TC-05	Logout	Login, then logout	Login page appears	Record after execution	Pending

Test Case Details

Field	Details
Test Case ID	TC-01 to TC-05
Module	Login / Authentication
Feature Under Test	Authentication and logout
Test Type	System Testing
Test Technique	Black Box Testing
Priority	High
Preconditions	Python, Chrome, Selenium installed and website accessible
Test Data	Valid, invalid, and empty credentials
Expected Result	Valid login succeeds; invalid login is rejected; logout returns to login page
Actual Result	Fill after actual execution
Status	Fill Pass/Fail after actual execution

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open SauceDemo website	https://www.saucedemo.com/	Login page displayed	Record after execution	Pending
2	Enter valid credentials	standard_user / secret_sauce	Products page displayed	Record after execution	Pending
3	Enter invalid password	standard_user / wrong_password	Error displayed; login denied	Record after execution	Pending
4	Enter invalid username	wrong_user / secret_sauce	Error displayed; login denied	Record after execution	Pending
5	Submit empty credentials	Blank / blank	Validation error displayed	Record after execution	Pending
6	Login then logout	Valid credentials	Login page displayed after logout	Record after execution	Pending

Web Element Identification

Element Name	Locator Type	Locator Value	Purpose
Username	ID	user-name	Enter username
Password	ID	password	Enter password
Login Button	ID	login-button	Submit credentials
Error Message	CSS Selector	h3[data-test='error']	Verify error
Menu Button	ID	react-burger-menu-btn	Open menu
Logout Link	ID	logout_sidebar_link	Log out

Assertion Validation Report

Assertion	Expected Result	Actual Result	Status
Page Title	Swag Labs	Record after execution	Pending
URL after valid login	Contains /inventory.html	Record after execution	Pending
Invalid credentials	Error message visible	Record after execution	Pending
Empty credentials	Validation error visible	Record after execution	Pending
Logout	Returns to login page	Record after execution	Pending

Exception Handling Report

Exception	Possible Cause	Handling / Remarks
NoSuchElementException	Element not found	Check locator and page state
TimeoutException	Element not visible in time	Use WebDriverWait
WebDriverException	Browser or driver issue	Check Chrome and Selenium setup
ElementNotInteractableException	Element not ready for input	Wait for visibility/clickability

Execution Summary

Metric	Value
Total Test Cases	5
Executed	Fill after execution
Passed	Fill after execution
Failed	Fill after execution
Execution Time	Fill after execution

Screenshot Checklist

No.	Screenshot	Filename	Capture Point	Status
1	Login page	01_login_page.png	After opening site	Pending
2	Successful login / Products page	02_valid_login_products.png	After valid login	Pending
3	Invalid password error	03_invalid_password_error.png	After submitting wrong password	Pending
4	Invalid username error	04_invalid_username_error.png	After submitting wrong username	Pending
5	Empty credentials error	05_empty_credentials_error.png	After empty submission	Pending
6	Logout result	06_logout_login_page.png	After logout	Pending
7	Terminal output	07_terminal_results.png	After script completes	Manual capture

Questionnaire

Question	Answer
What is authentication testing?	Testing that valid users can log in and invalid credentials are rejected.
What is an assertion?	A check that compares actual behavior with expected behavior.
What is exception handling?	Handling runtime errors safely in a program.
Which exception occurs when an element cannot be found?	NoSuchElementException.
Which method clicks the login button?	click().

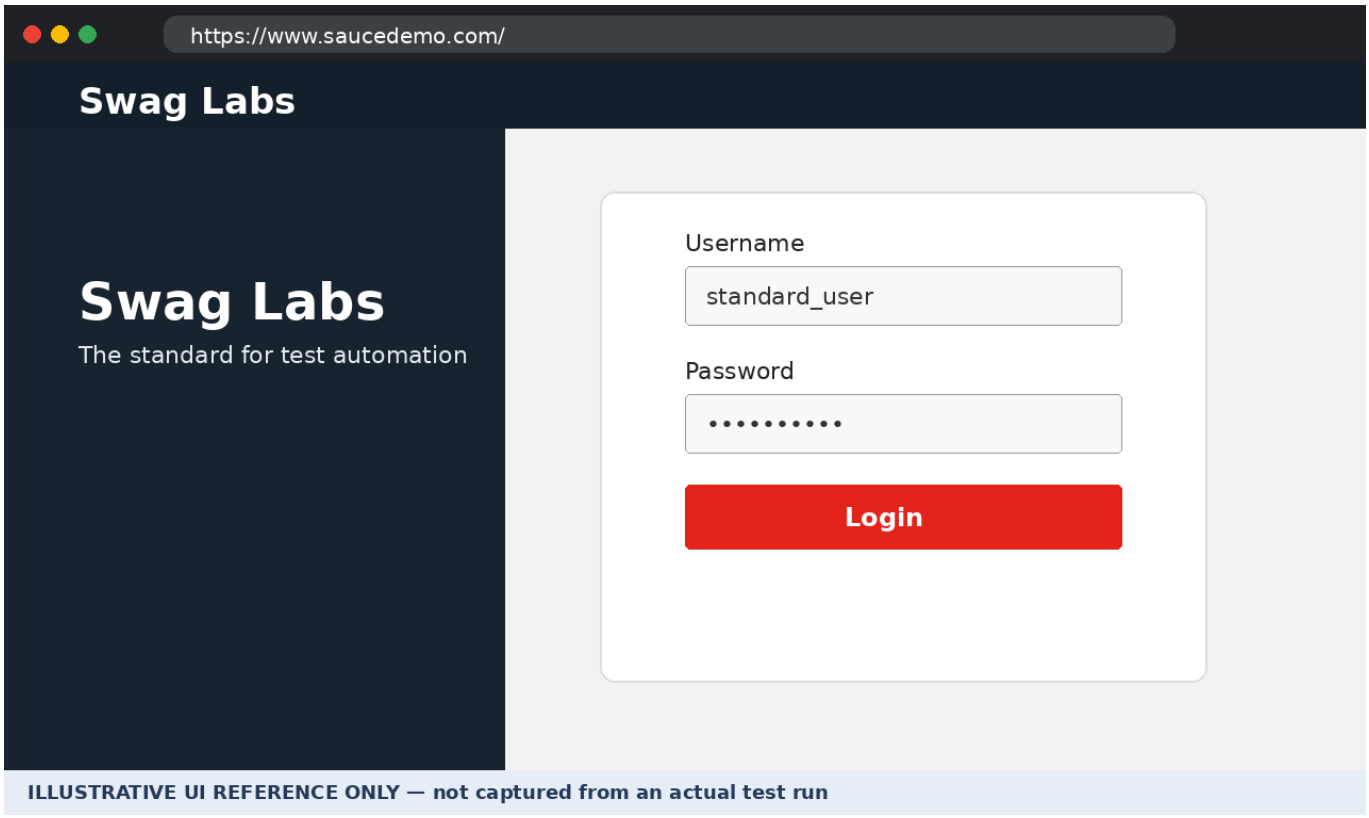
Conclusion

Conclusion / Outcome
A Selenium script is prepared to test valid login, invalid password, invalid username, empty credentials, and logout on SauceDemo. Record actual results after running the script.

SCREENSHOT REFERENCE PAGES

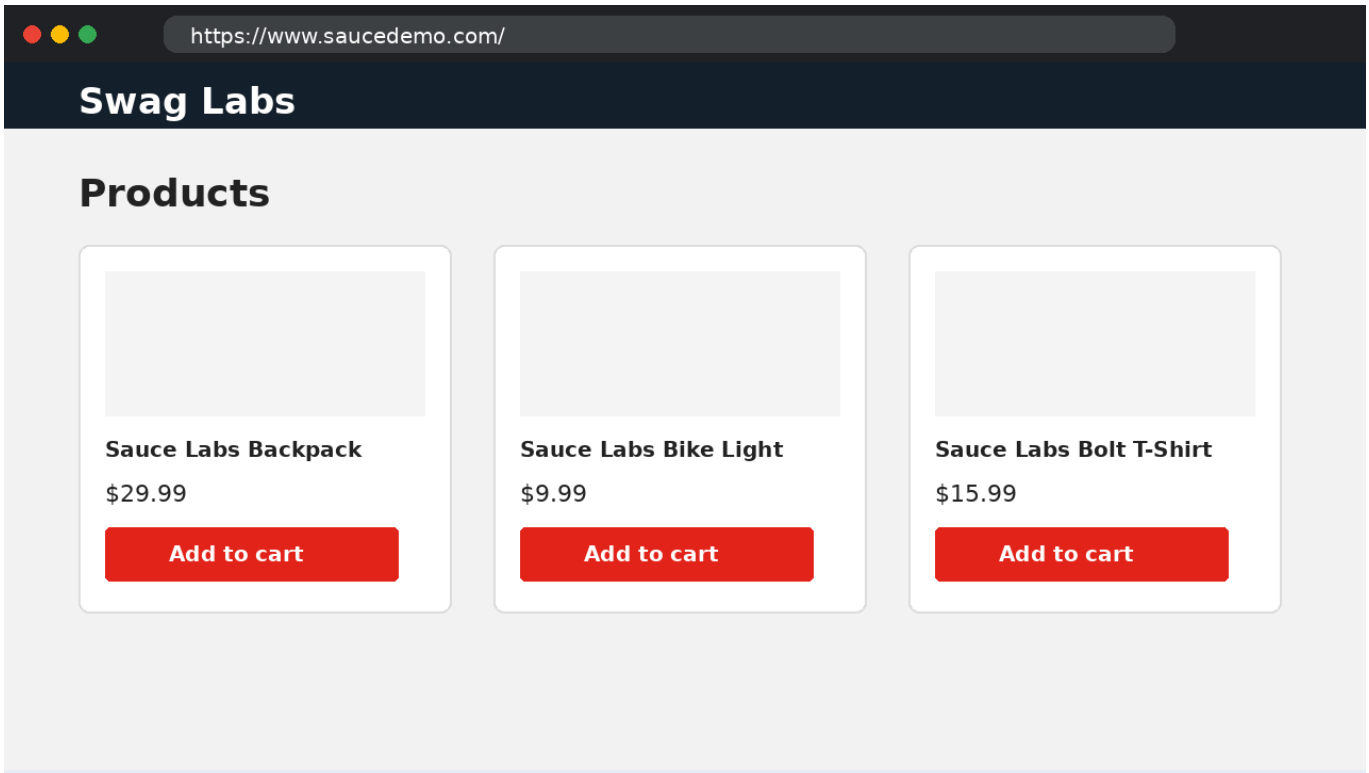
Note: The images below are illustrative UI references. They are not evidence of a real Selenium run. Run tutorial15_login_testing.py to capture actual screenshots in the Tutorial_15_Screenshots folder.

Login page reference



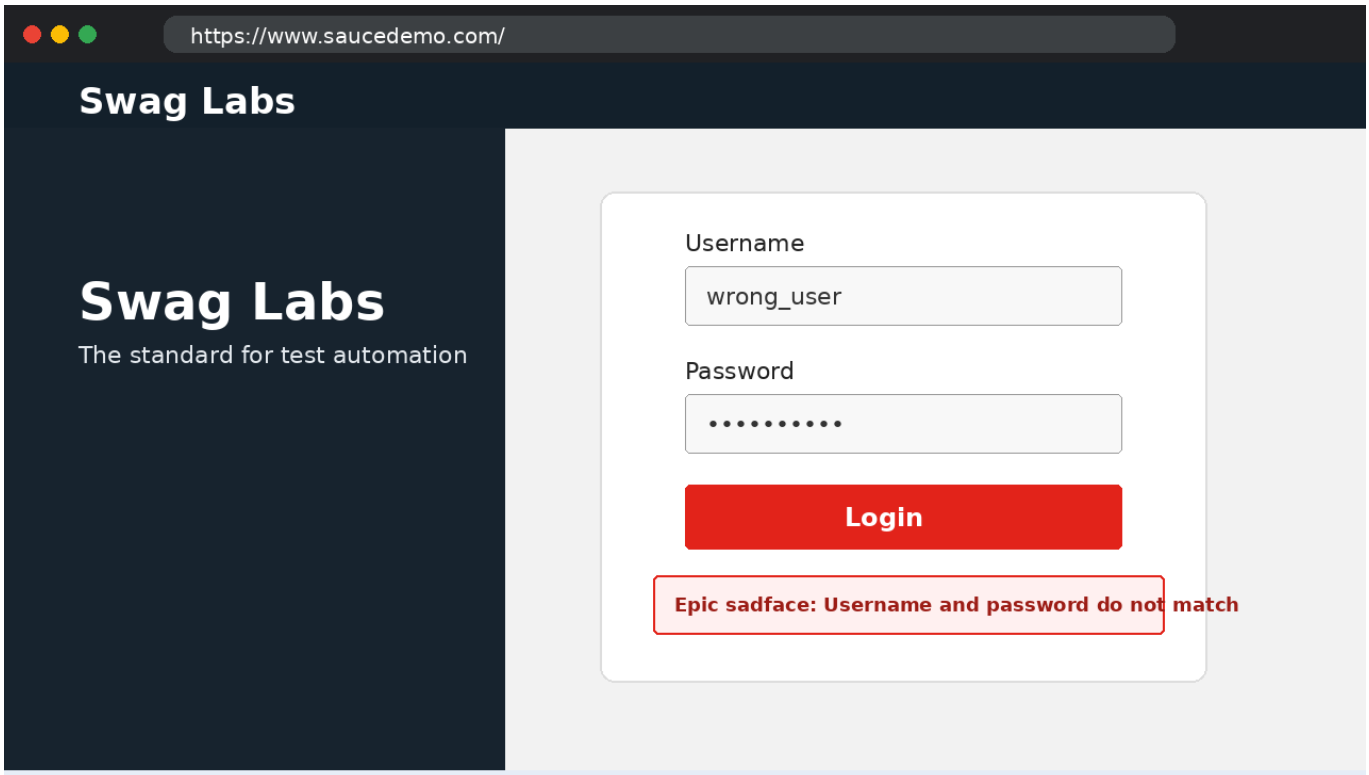
ILLUSTRATIVE UI REFERENCE ONLY — not captured from an actual test run

Successful login / Products page reference



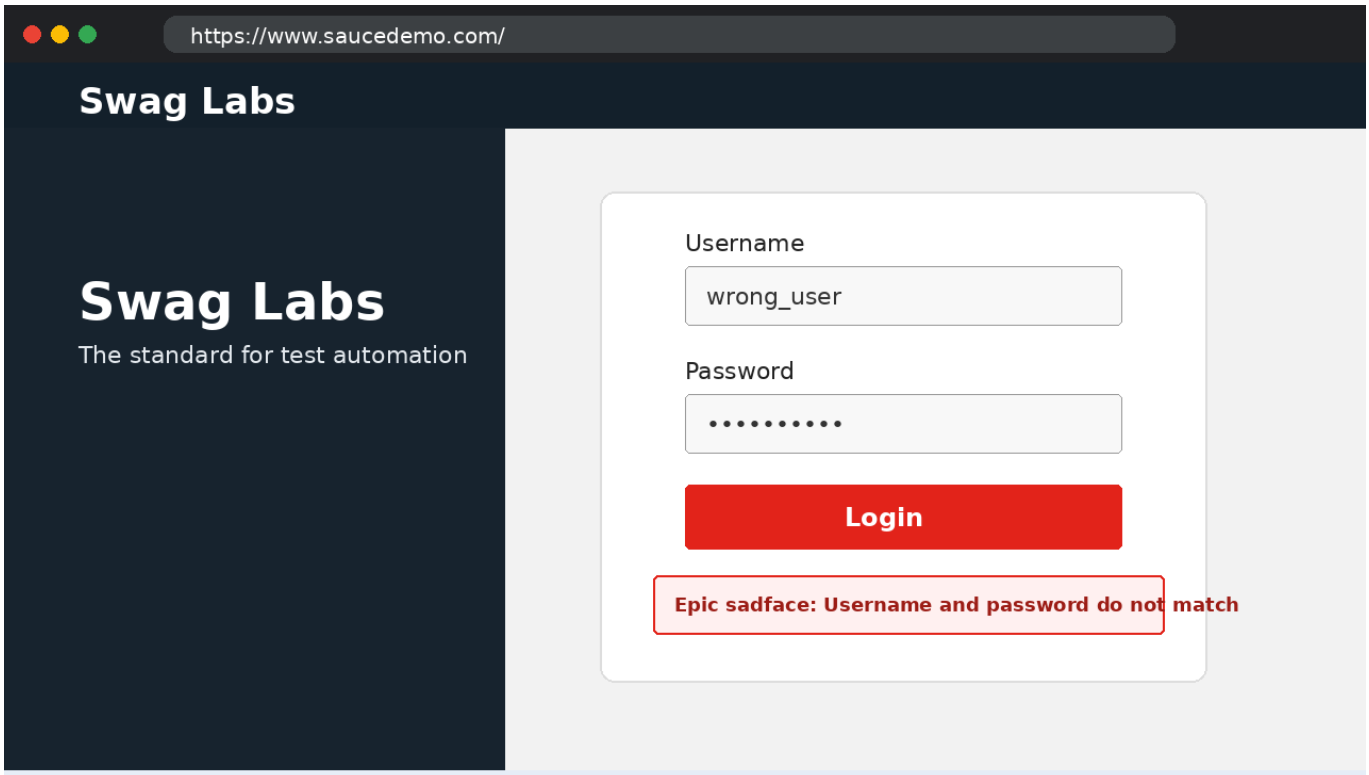
ILLUSTRATIVE UI REFERENCE ONLY — not captured from an actual test run

Invalid password error-state reference



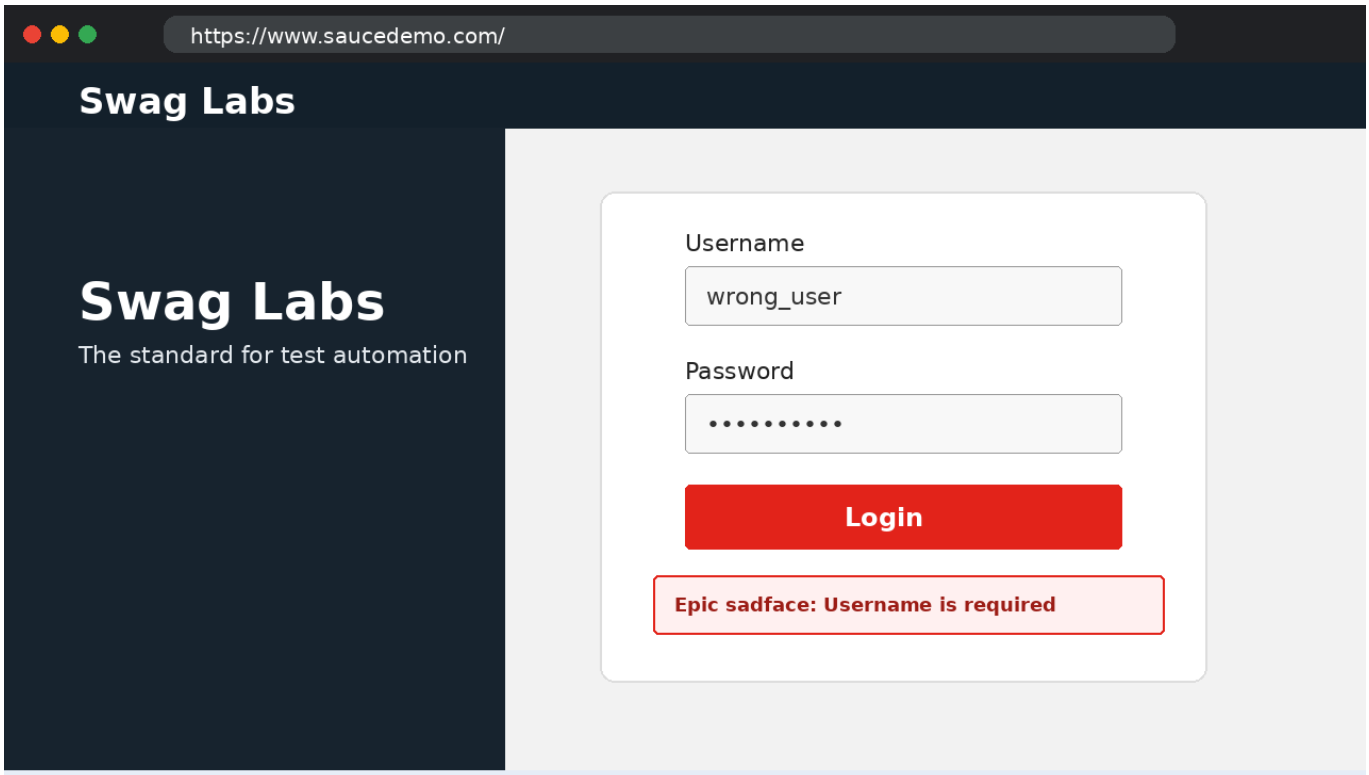
ILLUSTRATIVE UI REFERENCE ONLY — not captured from an actual test run

Invalid username error-state reference



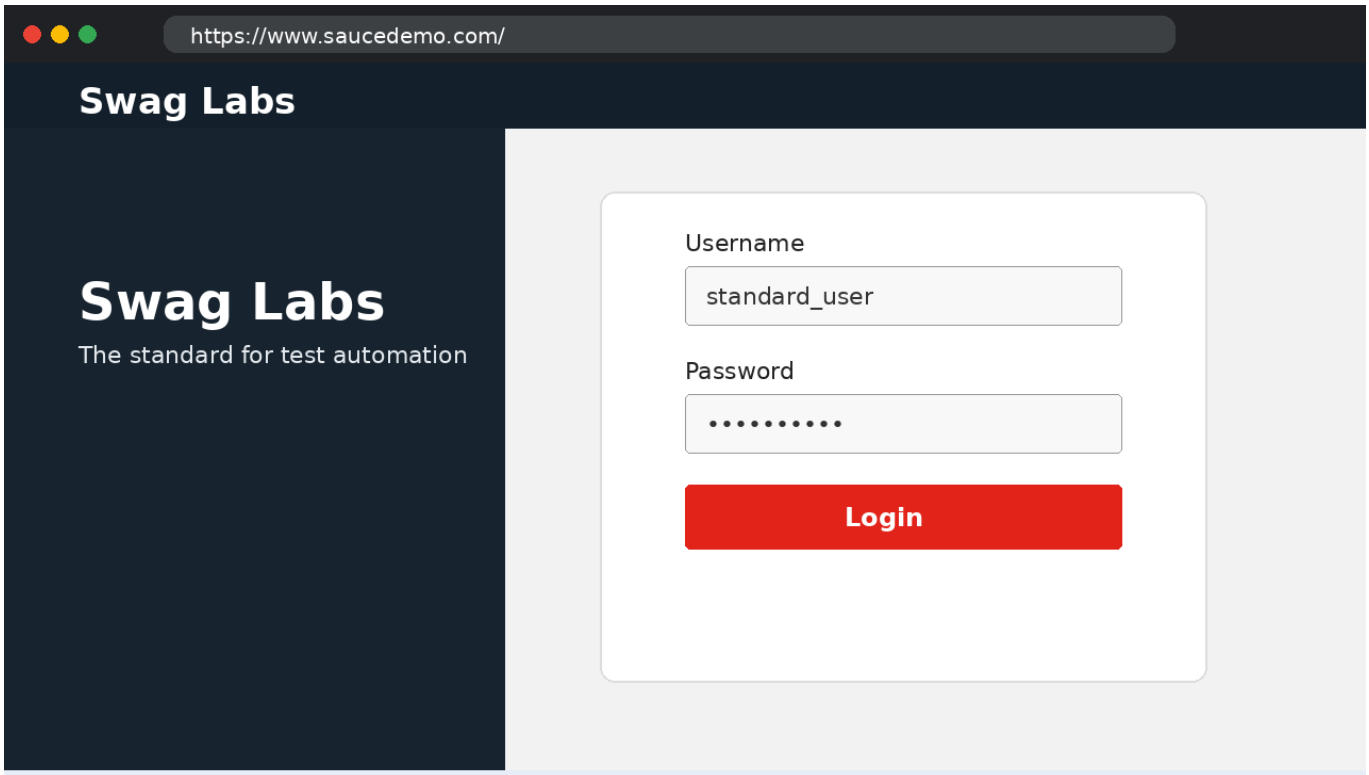
ILLUSTRATIVE UI REFERENCE ONLY — not captured from an actual test run

Empty credentials error-state reference



ILLUSTRATIVE UI REFERENCE ONLY — not captured from an actual test run

Logout returns to login page reference



ILLUSTRATIVE UI REFERENCE ONLY — not captured from an actual test run